BTCS301N

III Semester Examination, January - 2023

B.Tech./B.Tech.+M.B.A./B.Tech.+M.Tech.(CSE,CSE-CC,CSE-CF,CSE-BDA,IT,BDCE-Impetus,CSE-AIML,CSE-BDAI,CSE-CMCI,CSE-MA,CSE-ICS,CSE-AII,CSE-ES,CSE-DSI,CSE-FSDI)

Discrete Structures

Choice Based Credit System (CBCS)

Time: 3 Hrs. **Maximum Marks: 60**
Minimum Pass Marks: 24

Note: All questions carry equal marks, out of which part'A' and 'B' carry 3 marks and part 'C' carries 6 marks. From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice. Draw neat diagram, wherever necessary. Assume suitable data wherever necessary.

Q.1(A) Write the principle of inclusion and exclusion. 03

(B) Explain Partial Order relation with taking suitable example. 03

(C) What do you mean by Transitive of a relation? Check given relation R={(1,1) (2,1) (2,2) (2,3) (2,4) (3,1) (3,2) (3,3) (3,4)} for set A={1,2,3,4} is transitive or not with the help of Warshall's algorithm. 06

OR

Define function. How many Onto function are there from an n element (n>=2) set to a 2-element set?

Q.2(A) Explain Disjunctive Normal Form and Conjunctive Normal Form. 03

(B) Explain the different rules of inference with certain examples. 03

(C) Prove that following is Tautology or not:-

I. $(p \vee q \vee r) \leftrightarrow [((((p \rightarrow q) \rightarrow q) \rightarrow r) \rightarrow r]$

II. $(p \leftrightarrow q \wedge r) \Rightarrow (\neg r \rightarrow \neg p)$ 06

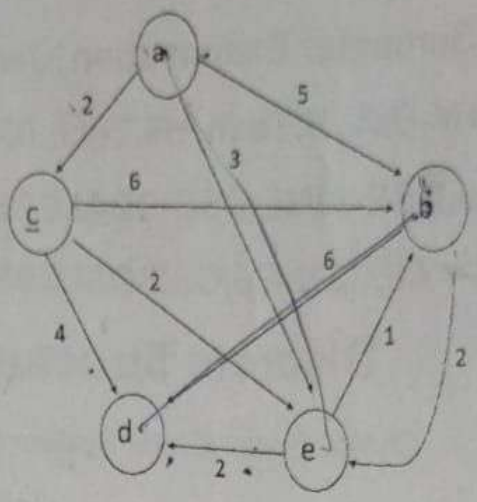
OR

Explain quantifier in predicate logic and its types with taking suitable example.

Q.3(A) Define Isomorphic Graph. How many simple non isomorphic graph are possible with 4 vertices and 3 edges. 03

(B) Explain different Tree traversal methods with taking suitable example. 03

(C) Find the shortest path between *a* and *d* for the following graph given in figure-1.



06

figure-1

OR

Find a minimum spanning tree for the weighted graph in figure-2 using Prim's algorithm?

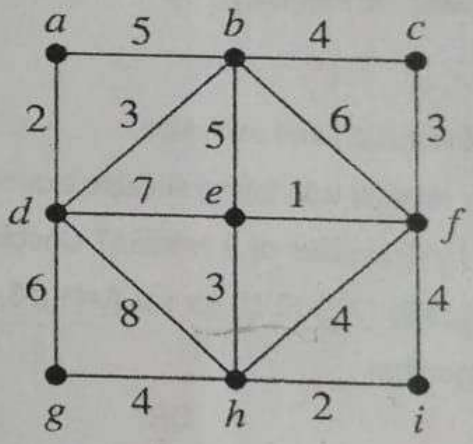


figure-2

Q.4(A) Discuss ring and field with example. 03

(B) What is Group? Explain the different properties of Group. 03

(C) Prove that (R, +) is an Abelian group, where R be the set of all real number and '+' be the ordinary addition operation. 06

OR

Prove that any group of order 3 is cyclic.

Q.5(A) Solve the recurrence relation f(n)=3f(n-1)+1. 03

(B) Describe the lattice and its properties. 03

(C) Let S= {a, b, c, d} and P(S) be its power set. Let ⊆ be the inclusion relation on P(S). Draw Hasse diagram. 06

OR

Using generating function, solve the recurrence relation $an = 3an-1$ for $n \geq 1$ with $a0=2$.